Weekly Report

Clustering Algorithm Work

Dean Reiter

TRIUMF DarkLight Group Meeting

July 14, 2026

Summary

1. Reviewed Andy's algorithm in more detail
2. Researching other peak detection and clustering algorithms
3. Working on code to test clustering algorithm

Reviewing Andy's Algorithm

- Don't completely understand "gaussian quality"
- Thresholds need to be lowered – algorithm tends to miss clusters
- Clusters seem visually indistinguishable from noise in cosmic data

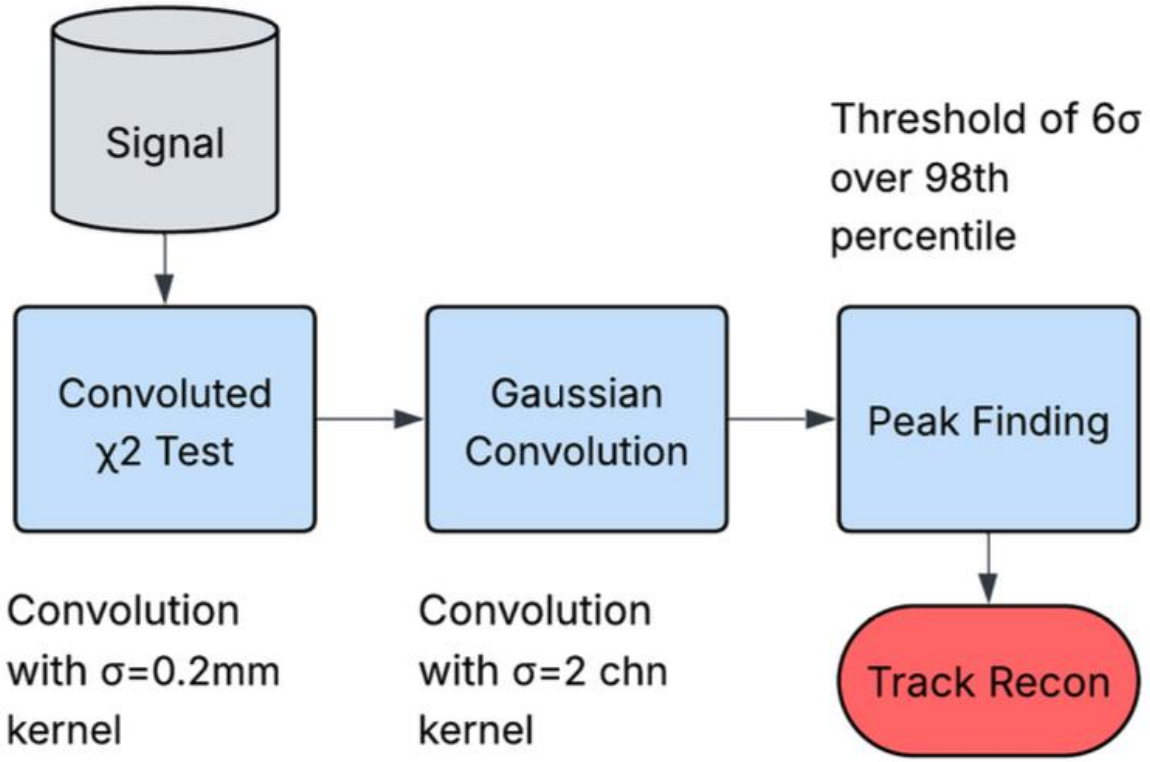


Figure 3.1: Pipline for GEM clustering

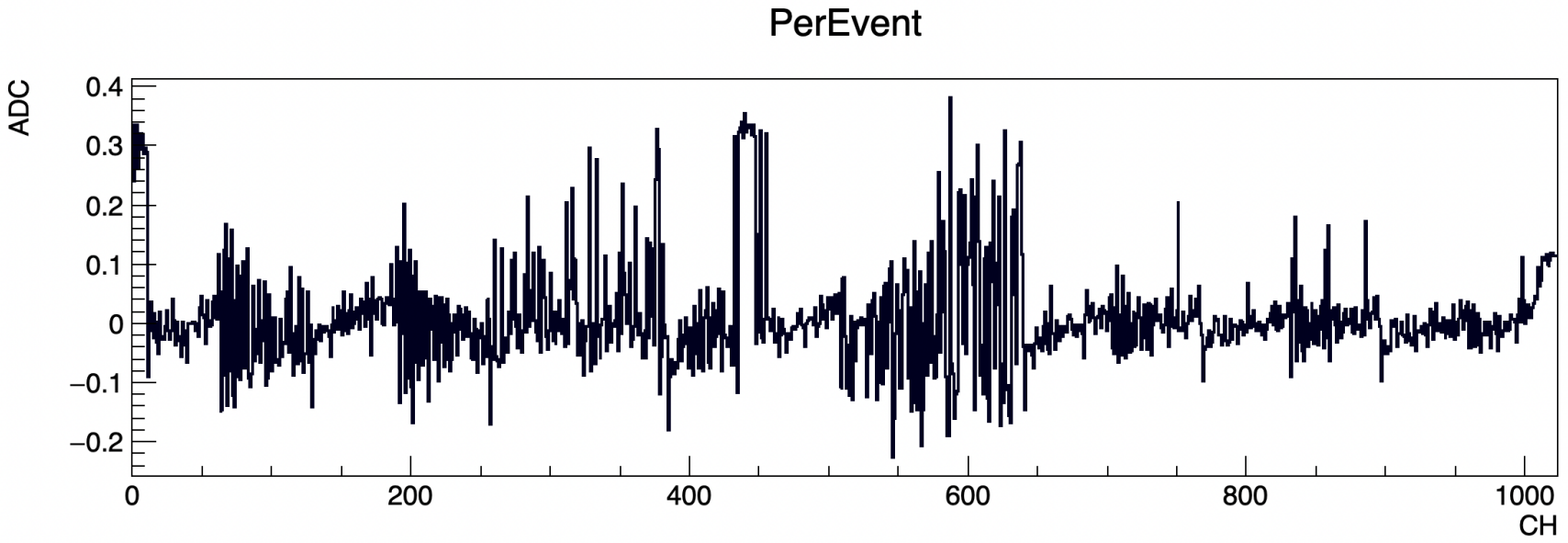
Credit: Andy Yang

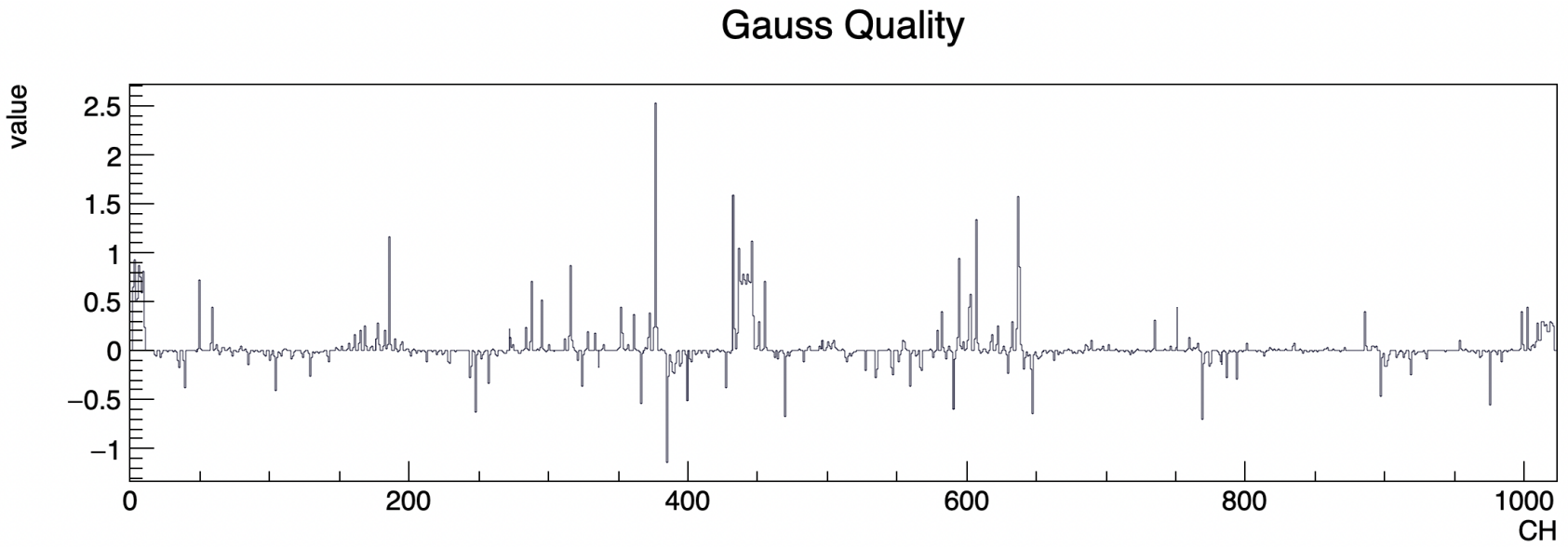
Convoluted χ^2 Test

Calculate “Gaussian quality” of each channel

How well window of five channels around each channel fits gaussian distribution with σ ≈ 1/2 channel width ≈ 0.2 mm

$$Q(i) = G(i)\,\frac{\chi^2_{\mathrm{NH}}(i) - \chi^2_{\mathrm{GH}}(i)}{\chi^2_{\mathrm{GH}}(i)}$$





Gaussian quality

Gaussian quality: $\mathrm{Q}(i) = \frac{(\overrightarrow{x_i}\cdot\hat{n})^3}{\overrightarrow{x_i}^2 - (\overrightarrow{x_i}\cdot\hat{n})^2}$

$\overrightarrow{x_i}$ = set of 5 adc values centered around channel i
$\hat{n}$ = set of 5 points in Gaussian distribution with σ ≈ 1/2 channel width, centered around zero, normalized

- Clusters have variable widths depending on angle, particularly in dispersive direction
- May be a simpler and faster way to quantify gaussian quality

Researching Other Clustering Algorithms

- Simple peak detection
  - Threshold + analyze first and second derivatives
- Identifying gaussian signals
  - Filters or convolutions e.g. matched filter
  - Basic fit – assume σ based on FWHM
- Reading papers from other experiments

Testing Clustering Algorithms

Working on code to test clustering algorithm

- Maximize signal-to-noise ratio (SNR) of identified clusters
- Maximize efficiencies
  - What fraction of clusters get matched to trigger hits?
  - What fraction of trigger hits get matched to tracks?
- Minimize computation time